



EE 574: Biomedical Signal and Image Analysis

Course Code	EE 574
Course Name	Biomedical Signal and Image Analysis
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Open Elective for All 3rd/4th year (odd-sem) BTech/MTech/MS/MSc/MA/Ph.D.
Mutual Exclusion	EEXXX (Signals & Systems)
Approval	54th BoA (May 2024)

Course Contents

- **Introduction to fundamental concepts:** Signal processing overview; Fundamental signals (1-D and 2-D); Classification of systems; Characteristics of linear and non-linear systems, Concepts of convolution and Autocorrelation. Frequency analysis: Fourier Transform, DTFT, FFT, Welch's method; Data Acquisition: Sampling in time, aliasing, interpolation, and quantization. Introduction to random variables and probability density functions (PDFs). Stationary and Nonstationary Processes. (4 Hours)
- **Electrophysiology and Electrographic modalities:** Electric properties of a cell: Ion transport, transmembrane potential, membrane resistance and capacitance, action potential, Hodgkin-Huxley model. Electric data acquisition and biomedical electrodes. ECG: Function and structure of heart, cardiac muscle and excitation process, origin of ECG, ECG electrode placement, noise sources, Modeling and representation of ECG, ECG and cardiovascular diseases. EEG: Neuron, scalp potentials, EEG recording systems, EEG rhythms, Evoked potentials, EEG artifacts, Disease of CNS and EEG. EMG: Muscle, motor unit, muscle contraction, muscle force, noise in EMG, Neuromuscular disease and EMG. (6 Hours)
- **Biosignal processing:** Feature extraction methods: Frequency domain (PSD, dominant frequency, spectral entropy), Non-linear features (Lyapunov exponents, fractal dimensions, complexity and mobility measures), entropy-based measures, cross-correlation and coherence, phase-based features (phase synchronization, phase coherence, phase-amplitude coupling), time-frequency methods (short-time Fourier Transform, Wigner-Ville distribution, Morlet Wavelets), connectivity analysis (graph theoretic measures, synchronization measures, Granger causality). Noise Removal: Removal of linear and nonlinear trends, Filtering techniques active and passive filters; Design of digital filters. Generalized Eigen Decomposition (GED) for denoising, Empirical mode decomposition (EMD), and blind source. (10 Hours)
- **Introduction to Medical Images:** Imaging Modalities: introduction of major modalities for medical imaging: ultrasound, X-ray, CT, MRI, PET, and SPECT. Image formation and tissue energy interaction as a basis for different imaging modalities, quantitative medicine, multi-scale and multi-resolution relations in medical diagnostics, Cardiovascular and Neuro Imaging and Analysis. Automated Image Quality Assessment in Medical Images. (8 Hours)
- **Statistical and Structural Analysis:** Statistical and structural classification, morphological and topological component analysis. Rigid and non-rigid transformations. Co-registration of images, and motion correction. Extension of filtering and Fourier methods to 2-D signals and systems. Noise reduction methods, optimal and adaptive filters, homomorphic filtering, edge detection, time-frequency, and fractal analysis. Bayesian graphical modeling and inference for image restoration. (8 Hours)
- **Pattern Analysis:** Pattern classification and diagnostic decision-Measures of diagnostic accuracy. Abnormality detection and pattern recognition in cardiac, brain, and abdominal images, image categorization, and computer-aided diagnosis. Image descriptors, Image similarity, rendering surfaces and volumes, 3D neurostructure analysis, hypothesis testing, and prediction modeling. Deep learning for medical images. (6 Hours)

Laboratory Modules

- There shall be practical demonstration of certain fundamental algorithms of data processing and its analysis that shall be explained in computational modeling fashion. This shall be conducted in the form of minor projects and analyzing real problems in clinical practice to aid theoretical concepts.

Text Books

1. John L. Semmlow, **Biosignal and Biomedical Image Processing: MATLAB Based Applications**, CRC press.

References

1. Rangayyan R M, **Biomedical Image Analysis**, 5th Edition, CRC Press, 2005
2. Atam Dhawan, **Medical Image Analysis**, 2nd edition, by WILEY.
3. Reddy, D.C., **Biomedical signal processing: principles and techniques**, McGraw-Hill, 2005.
4. Gonzalez, R., and R. E. Woods., **Digital Image Processing**, 2nd Edition, Prentice-Hall, 2002.
5. E.N. Bruce, **Biomedical Signal Processing and Signal Modelling**, John Wiley and Sons, 2001.